AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 7 · STUDENT LESSON

Parameters and architecture

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

In the Week 6 network, which quantities change when x changes while the model stays fixed?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Distinguish architectural choices from parameter values. Try the recall before opening notes.

Your learning target

I can distinguish architectural choices from **parameter** values.

I can count weights and biases in a small dense network.

I can explain why **parameter** count alone cannot establish quality.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Distinguish architectural choices from **parameter** values

Count weights and biases in a small dense network

Explain why **parameter** count alone cannot establish quality

Retrieve what you already know

> In the Week 6 network, which quantities change when x changes while the model stays fixed?

My starting explanation

This week's end goal

By the end, I can explain **parameter**s and **architecture** using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Parameters and architecture case. Separate observations, given facts and assumptions before choosing an explanation.

Consider a dense network with 2 **input**s, 3 hidden units using ReLU, and 1 **output** unit using identity. We call its widths 2 to 3 to 1. These widths, connectivity and activation choices describe its **architecture**.

The weights and biases supply the numerical values used by that **architecture**.

A fully connected hidden unit receiving 2 **input**s needs 2 weights plus 1 bias. Three such units need 6 weights and 3 biases. The **output** receives 3 activations and needs 3 weights plus 1 bias. Total: 13 **parameter**s.

Input values and computed activations are not additional **parameter**s.

Layer	Weights	Biases	Total
2 inputs to 3 hidden units	2 times 3 = 6	3	9
3 hidden values to 1 output	3 times 1 = 3	1	4
Whole network	9	4	13

This counting rule assumes one separate bias per **output** unit and no shared or tied weights. Other **architecture**s can use different rules.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Parameters and architecture. Connect each labelled input, deciding step and result.

For a **dense layer** with n **input** components and m **output** units, each **output** has n weights and one bias. The count is n times m plus m, or (n + 1) times m. Count each layer separately before adding.

```
def dense_count(inputs, outputs):
    return inputs * outputs + outputs

print(dense_count(2, 3))   # 9
print(dense_count(3, 1))   # 4
print(dense_count(2, 3) + dense_count(3, 1))  # 13
```

The helper assumes positive integer widths and the bias convention stated above. It is a counting aid, not a model trainer. A ReLU has no fitted numerical **parameter**s in this example.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Parameters and architecture. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Increase hidden width from 3 to4.

Predicted result

Prediction: First layer 12 **parameter**s; **output** layer 5; total 17.

Why the result follows

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Parameters and architecture to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Parameter	A numerical model value that a fitting procedure may adjust

Word	Meaning
Architecture	The arrangement of layers, connections and activation choices
Dense layer	A layer in which each output unit has a weight for every input component
Hyperparameter	A setting chosen for a model or training procedure rather than fitted as an ordinary weight

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Parameters and architecture, predict a step, then compare with its explanation.

Decide what a change alters

PROPOSED CHANGE	TYPE OF CHANGE	CONSEQUENCE
Change one weight 2 to 3	Parameter value	Same architecture
Use 4 hidden units	Architecture width	Different parameter count
Use learning rate 0.1	Training setting	No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

For a dense 2 to 4 to 1 network with one bias per unit, count each layer and the total. How many more parameters does it have than 2 to 3 to 1?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Parameters and architecture

Count trainable weights and biases separately from activation functions.

The facts we will use

Supplied facts

Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/7

Trainable parameter count

	weights	biases	total
input to hidden	6	3	9
hidden to output	3	1	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/7

Parameters and architecture

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count first layer

2 x 3=6 weights plus 3 biases gives 9 parameters.





2. Count output layer

3 x 1=3 weights plus 1 bias gives 4 parameters.





3. Total

Total trainable parameters=9+4=13; ReLU adds no parameters here.

CHANGE ONE THING

Increase hidden width from 3 to4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count first layer	2×3=6 weights plus 3 biases gives 9 parameters.
2. Count output layer	3×1=3 weights plus 1 bias gives 4 parameters.
3. Total	Total trainable parameters=9+4=13; ReLU adds no parameters here.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections.

What this example does not establish

Keep the stated limit

Parameter count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

Keep the stated limit

Read the labelled picture: How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Increase hidden width from 3 to4. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Parameters and architecture claim and explain a repair using the supplied evidence.

Two networks can share an **architecture** and still make different **prediction**s because their **parameter** values differ. Conversely, two **architecture**s can happen to produce identical **output**s on the few cases tested.

A count describes the model, not how well it has learned.

Check the claim against the facts

When comparing widths, keep the task, feature schema, **training** data, validation cases and evaluation rule fixed. Record the **training** budget and seed as well.

More **parameter**s can increase flexibility, but they can also increase resource use and over**fitting** risk. Neither outcome follows from counting alone.

Locate the earliest unsupported step

Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Parameters and architecture with support, then explain the deciding step in your own words.

A **dense layer** has 4 **input**s and 2 **output**s with one bias per **output**. Count weights, biases and total.

For a dense 2 to 4 to 1 network with one bias per unit, count each layer and the total. How many more parameters does it have than 2 to 3 to 1?

Explain your prediction to a partner before comparing results.

Practise with less help: Sort the quantities

Use workbook W2 for the connected example "**Parameter**s and **architecture**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Parameter**s and **architecture**. Proposed explanation: "Count one bias per **input**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Parameters and architecture. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

```
def dense_count(inputs, outputs):
    return inputs * outputs + outputs

print(dense_count(2, 3))   # 9
print(dense_count(3, 1))   # 4
print(dense_count(2, 3) + dense_count(3, 1))  # 13
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Parameters and architecture. Explain why each step is justified.

Explain → apply → challenge

A **dense layer** has 4 **input**s and 2 **output**s with one bias per **output**. Count weights, biases and total.

Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

A team says its 26-**parameter** model must outperform a 14-**parameter** model. Give a fair check and one reason the claim is unsupported.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **parameter**s and **architecture evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish architectural choices from **parameter** values.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

A reproducible description needs ordered **input** features, layer widths, connections, activations and the exact **parameter** values or a way to recover them.

A description of **training** also needs the data split and **training** settings. The label small neural network does not specify a computation.

> **Two teams both report 13 parameters. What would you request before claiming they built the same model?**

Next week defines a loss so that we can measure how a particular set of **parameter**s performs on labelled examples.

A team says its 26-**parameter** model must outperform a 14-**parameter** model. Give a fair check and one reason the claim is unsupported.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Parameters and architecture case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Decide what a change alters

PROPOSED CHANGE	TYPE OF CHANGE	CONSEQUENCE
Change one weight 2 to 3	Parameter value	Same architecture
Use 4 hidden units	Architecture width	Different parameter count
Use learning rate 0.1	Training setting	No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Count a new architecture independently

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Parameters and architecture, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Count trainable weights and biases separately from activation functions.

By the end, I can	How I will show it
Distinguish architectural choices from parameter values	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Count weights and biases in a small dense network	Explain the deciding step, show a trace or test, and state one limit.
Explain why parameter count alone cannot establish quality	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Parameters and architecture without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish architectural choices from **parameter** values

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Count trainable weights and biases separately from activation functions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.

1. Total

2. Count **output** layer

3. Count first layer

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Total Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Count one weight for each **input**-to-unit connection and one bias for each receiving unit.

Two **input**s connecting to three hidden units make six weights. The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase hidden width from 3 to4. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. First layer 12 **parameter**s; **output** layer 5; total 17.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: First layer 12 **parameter**s; **output** layer 5; total 17. Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Repair a missing building block

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original **input**s cannot be distinguished by this network's final **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

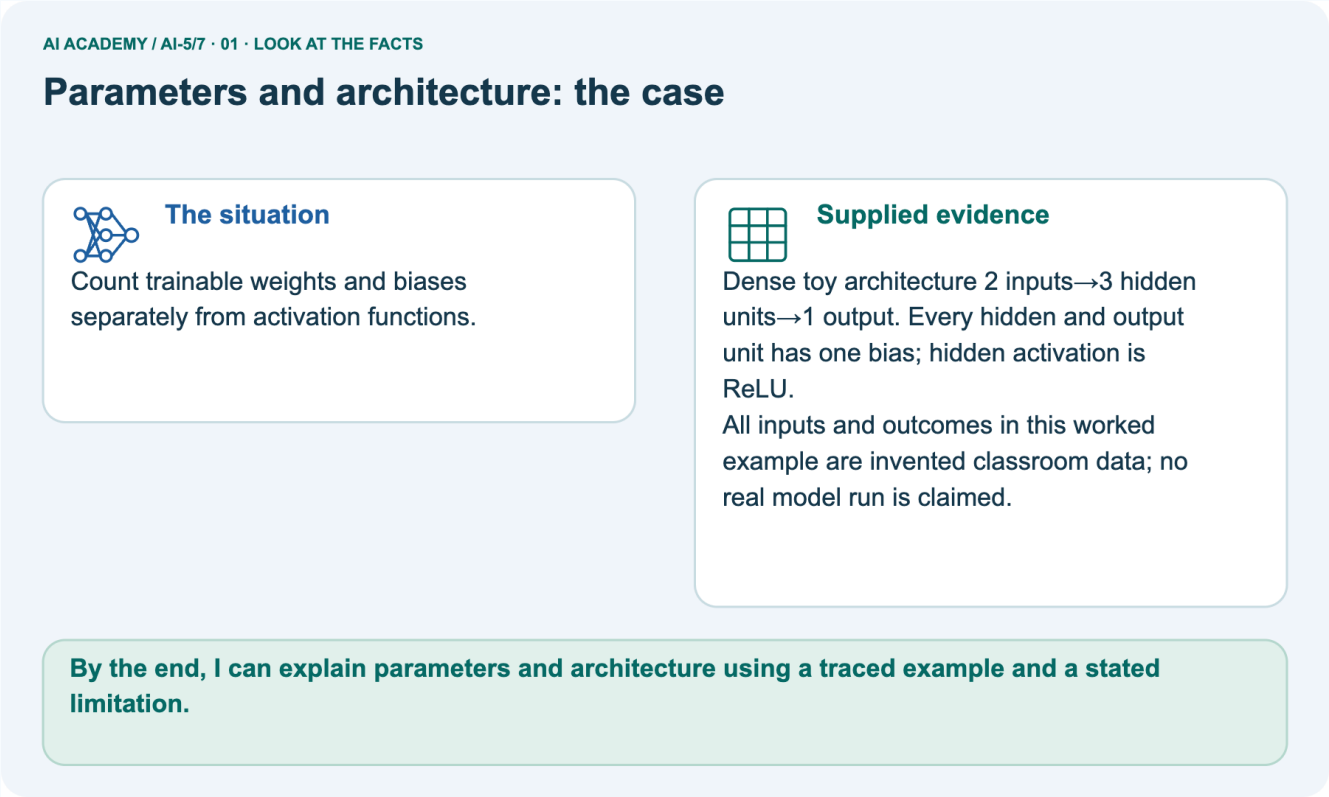
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Parameters and architecture graphic using its labels, deciding evidence and relevant limitation.

01. Parameters and architecture: the case



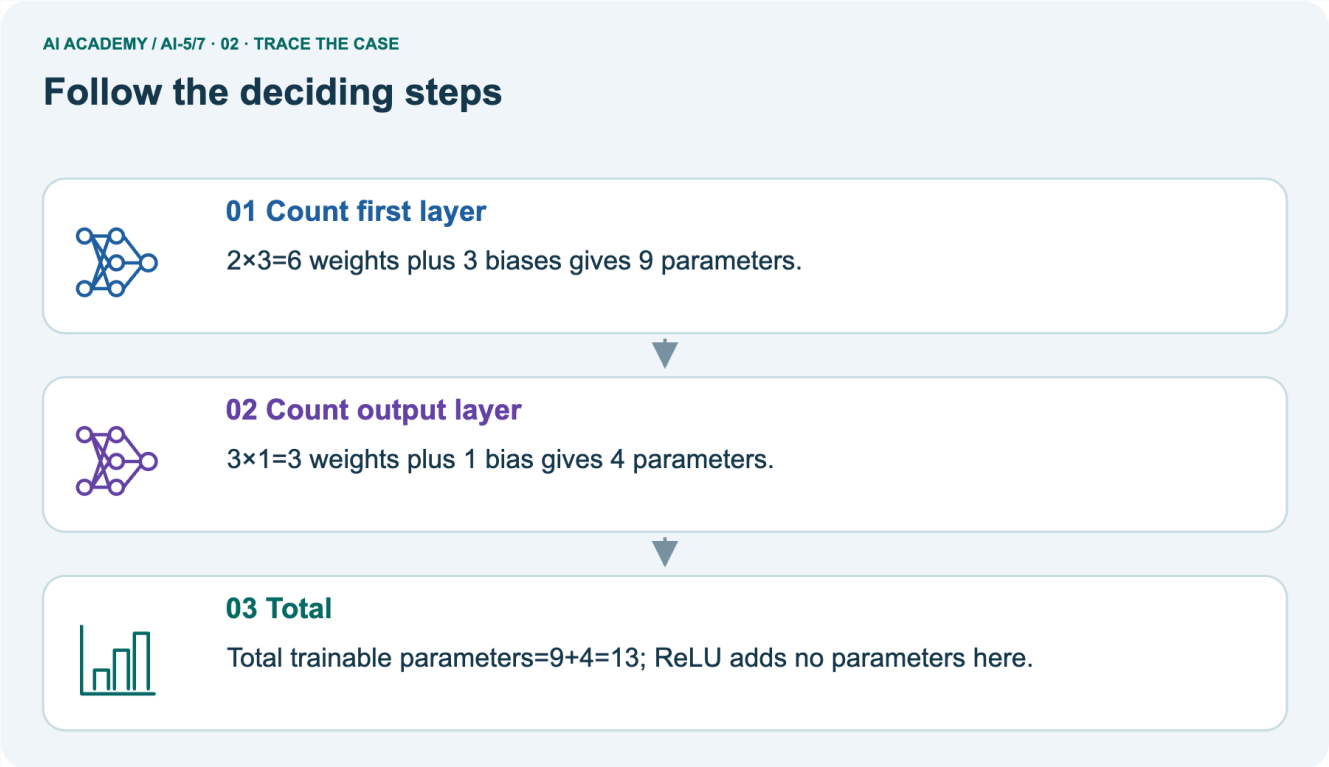
Follow the labelled picture with the matching explanation. Open the vector diagram

Count trainable weights and biases separately from activation functions. Supplied facts: Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

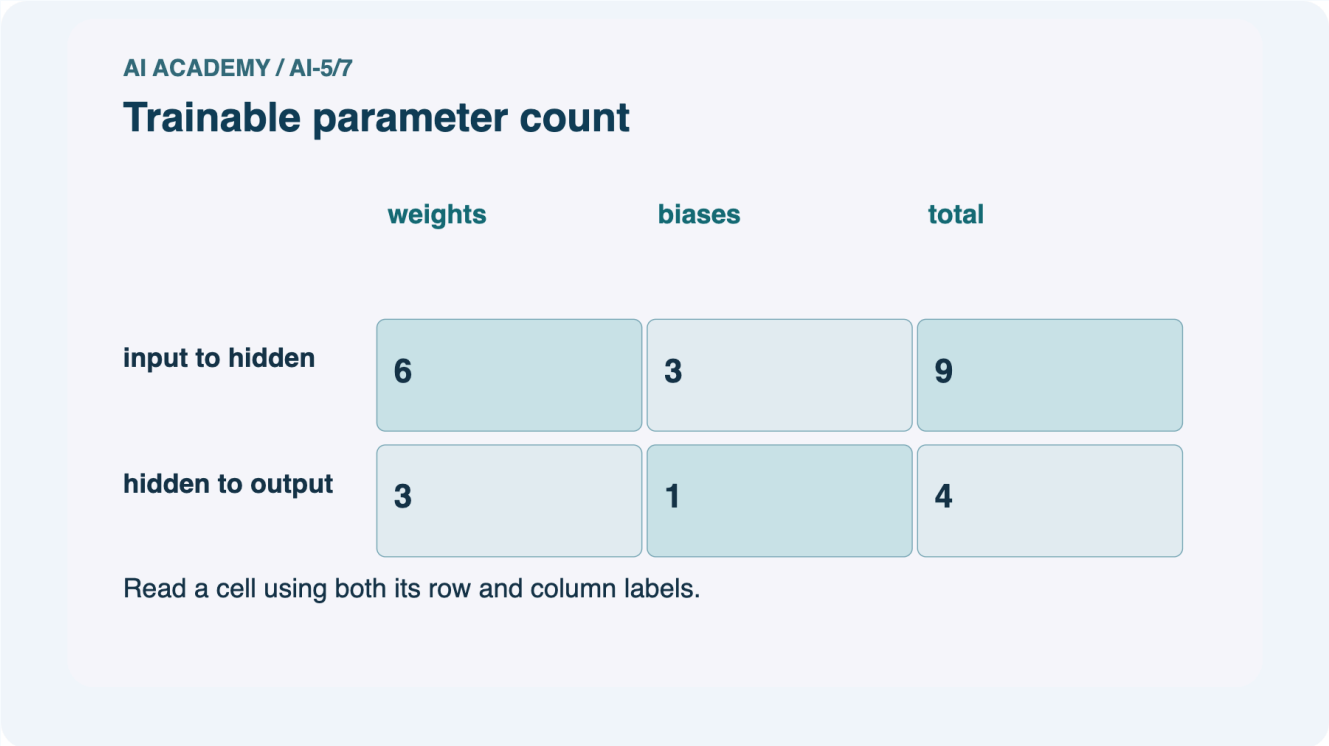


Follow the labelled picture with the matching explanation. Open the vector diagram

Count first layer: 2×3=6 weights plus 3 biases gives 9 **parameter**s. → Count **output** layer: 3×1=3 weights plus 1 bias gives 4 **parameter**s. → Total: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Trainable parameter count

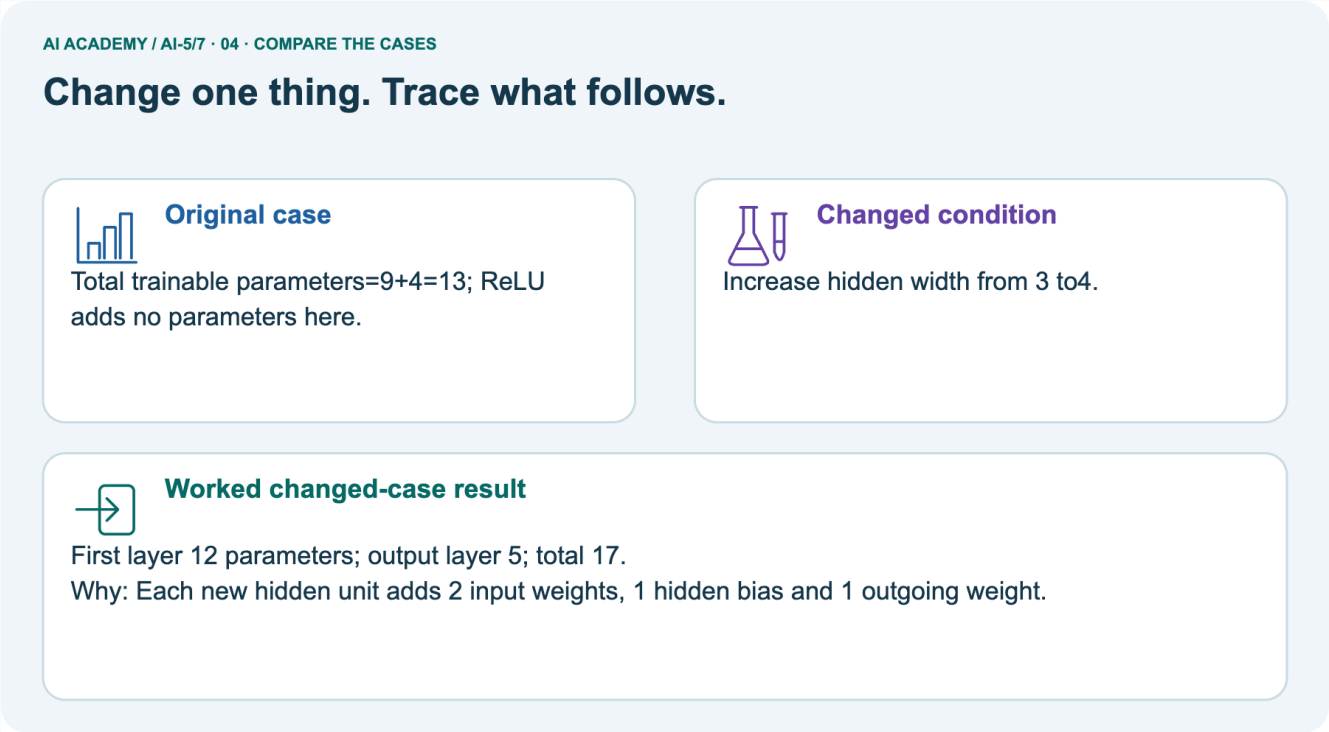


Follow the labelled picture with the matching explanation. Open the vector diagram

Trainable **parameter** count. **input** to hidden: weights = 6, biases = 3, total = 9. hidden to **output**: weights = 3, biases = 1, total = 4.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



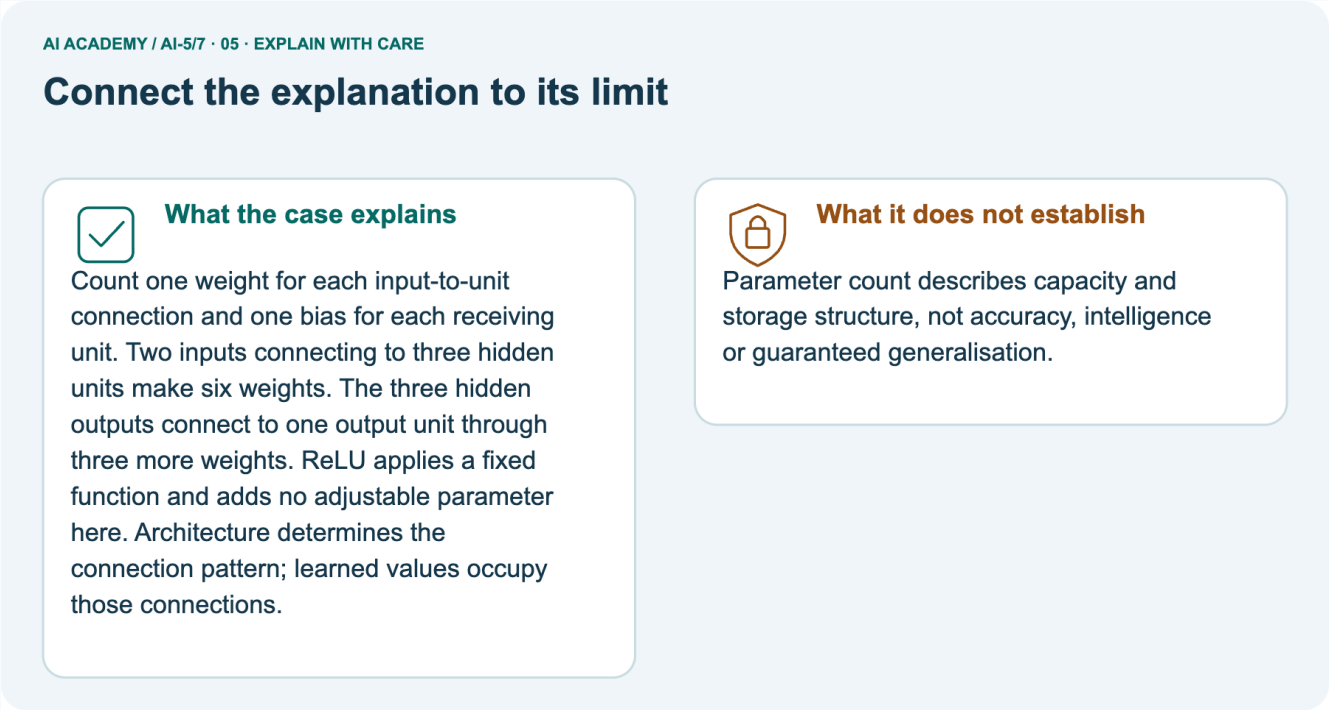
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Changed condition: Increase hidden width from 3 to4. Expected result: First layer 12 **parameter**s; **output** layer 5; total 17.

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

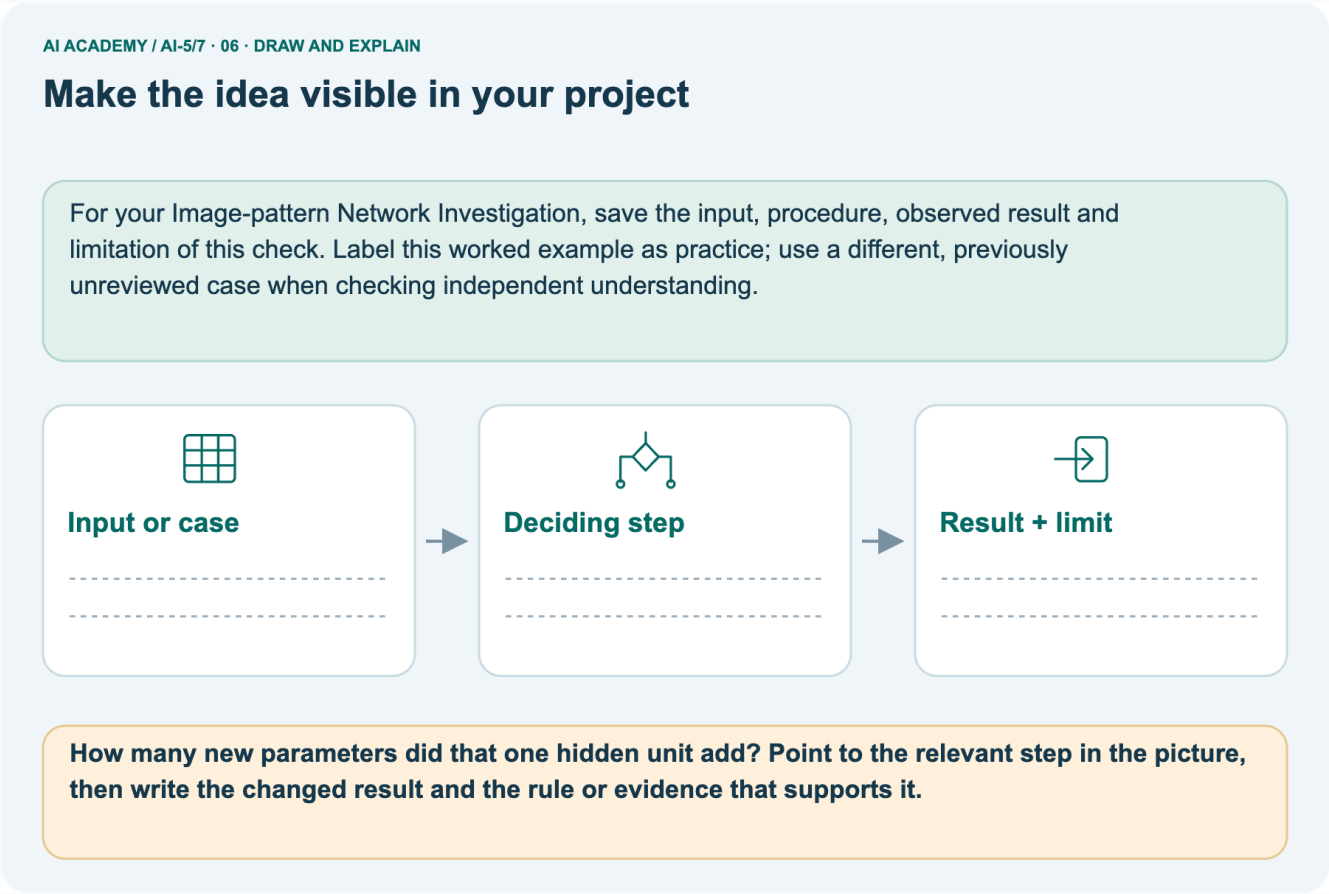
Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections. Limit: **Parameter** count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.